Topic 8

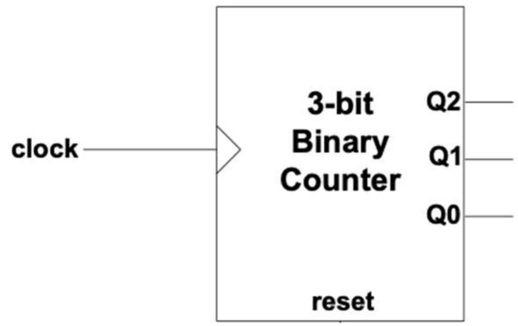
Counters

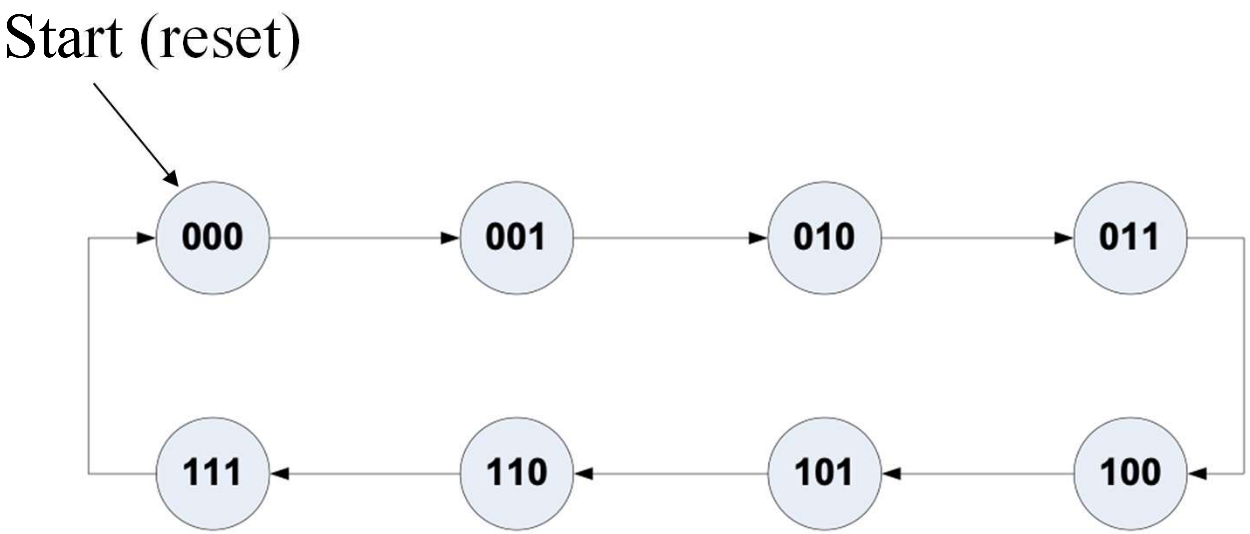
Counters

- A digital circuit that counts in different format: binary, decimal, one-hot, signed, unsigned…
- An *n*-bit binary counter can count up to $\boxed{}$ numbers
- An *n*-bit binary counter consists of $\boxed{}$ flip-flops
- Count up or count down, increment or decrement once per clock cycle – counting number of clock cycles
- Counters:
  - Asynchronous counters
  - Synchronous counters

Synchronous Binary Counter

- All the flip-flops are triggered by (synchronized to) the same – synchronous counter
- May be implemented by different type of flip-flops
- Example: a 3-bit binary counter can count through this sequence





Synchronous Binary Counter Design

- Following the counting sequence

Current Value (state)			Next Value (state)		
Q2	Q1	Q0	$Q2^+$	$Q1^+$	$Q0^+$
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0

Counter Implemented with D Flip-Flop

- Use D flip flops to hold values: $Q^+ =$ $\boxed{}$

Present State			Next State			D flip flop input		
Q2	Q1	Q0	$Q2^+$	$Q1^+$	$Q0^+$	D2	D1	D0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	0
0	1	0	0	1	1	0	1	1
0	1	1	1	0	0	1	0	0
1	0	0	1	0	1	1	0	1
1	0	1	1	1	0	1	1	0
1	1	0	1	1	1	1	1	1
1	1	1	0	0	0	0	0	0

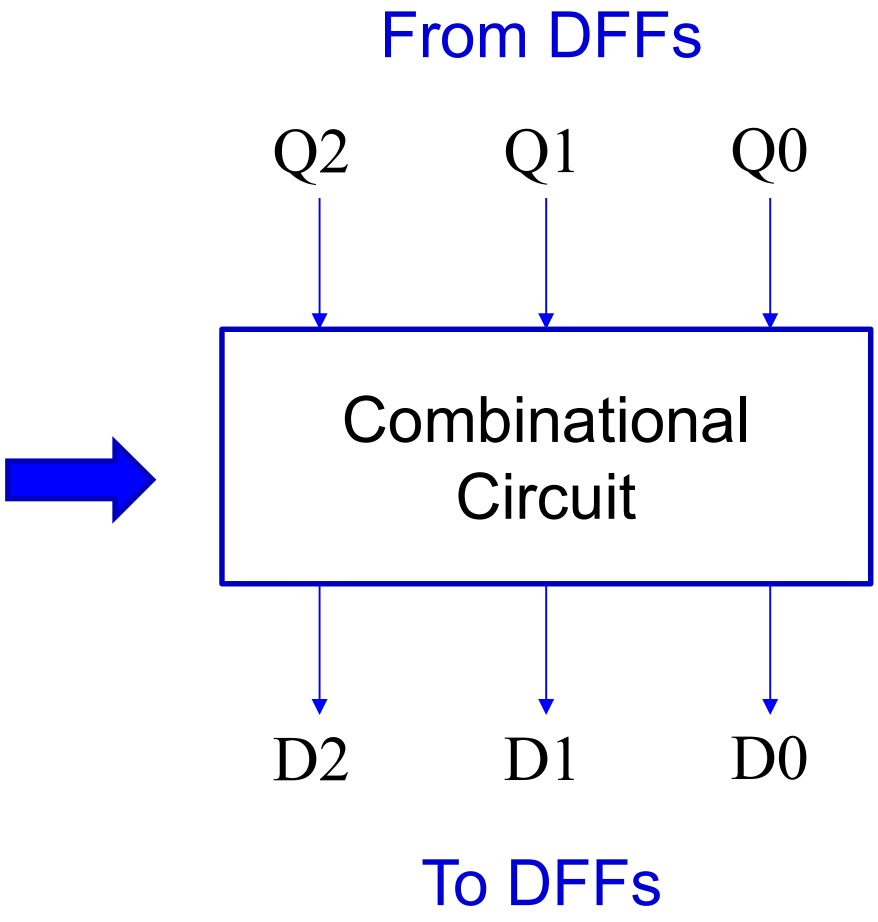
Same

Counter Implemented with D Flip-Flop

- Drop the columns for Q^+

Present State (D-FF outputs)			Next State (D-FF inputs)		
Q2	Q1	Q0	D2	D1	D0
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0

Truth table inputs   Truth table outputs



State Registers Implemented with D Flip-Flop

D2

Q2 \ Q1Q0	00	01	11	10
0	0	0	1	0
1	1	1	0	1

D1

Q2 \ Q1Q0	00	01	11	10
0	0	1	0	1
1	0	1	0	1

D0

Q2 \ Q1Q0	00	01	11	10
0	1	0	0	1
1	1	0	0	1

D2 =
=
=
=

D1 =
=

D0 =

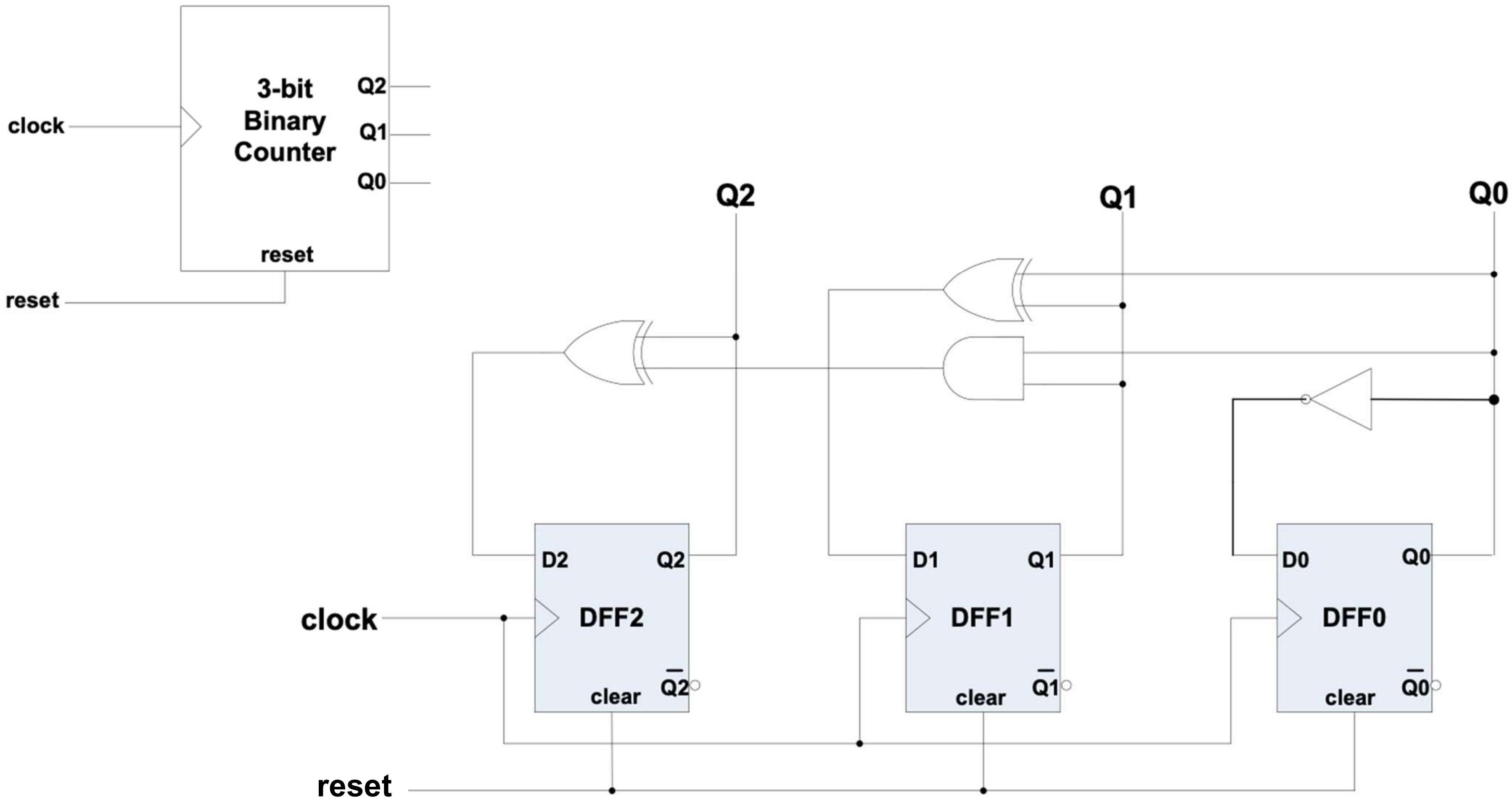
Present State			D flip flop input		
Q2	Q1	Q0	D2	D1	D0
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0

- The input equation can be generalized as

Dn =

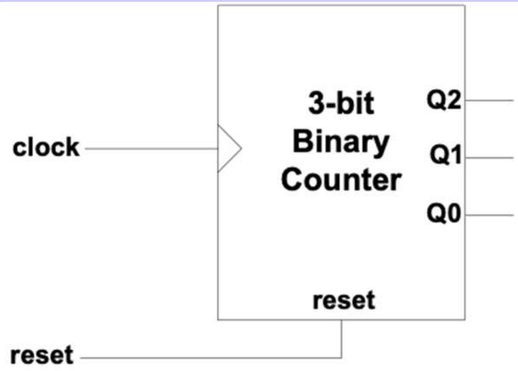
Synchronous Binary Counter with D Flip-Flop

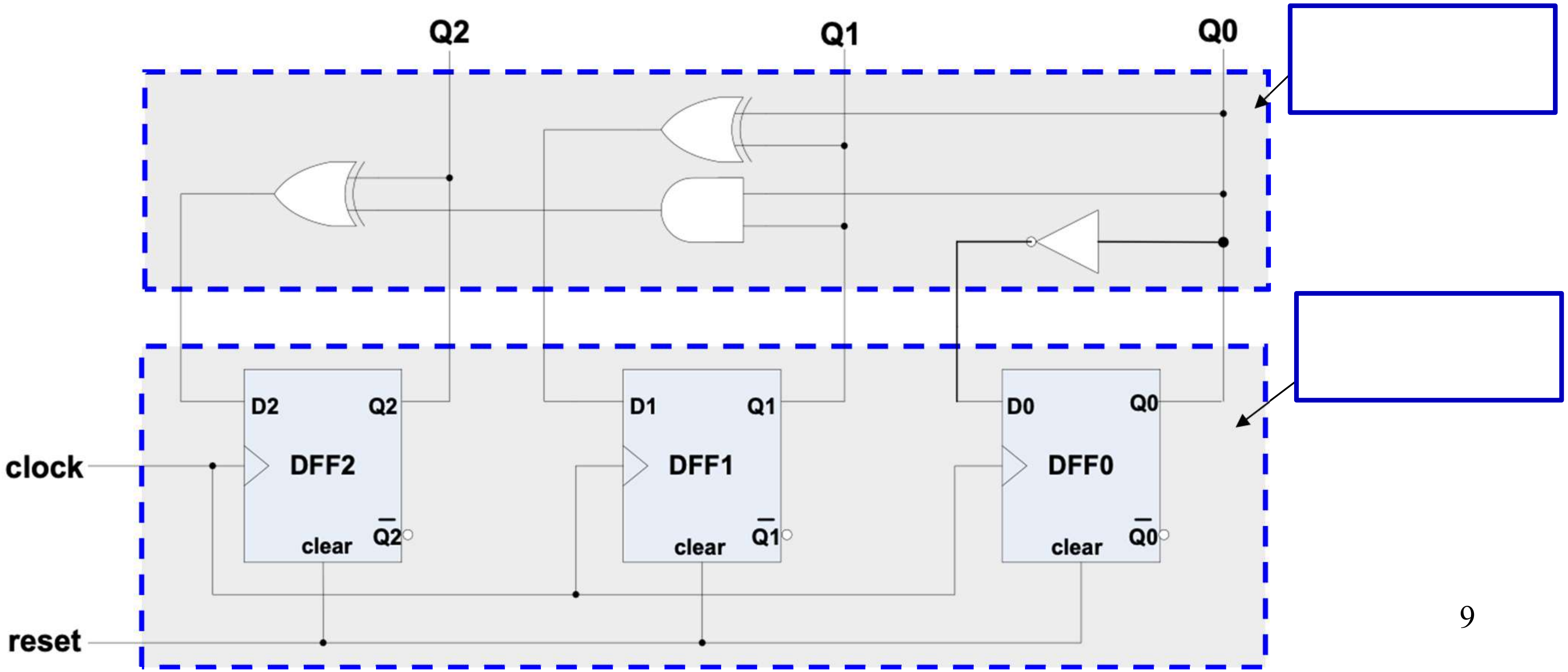
- 3-bit binary counter by D FF



Synchronous Binary Counter with D Flip-Flop

- 3-bit binary counter by D FF

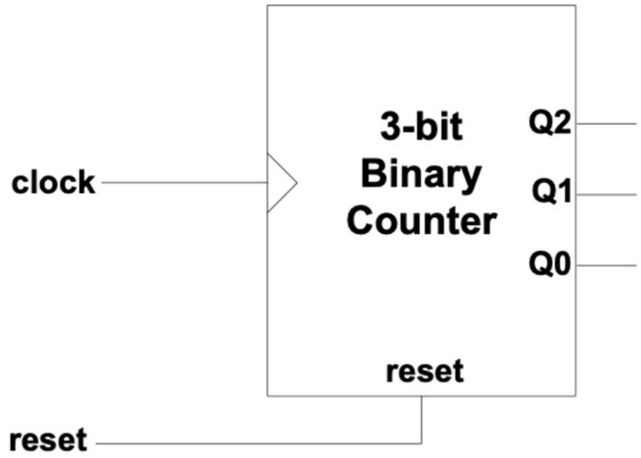




Verilog Model: Synchronous Binary Counter

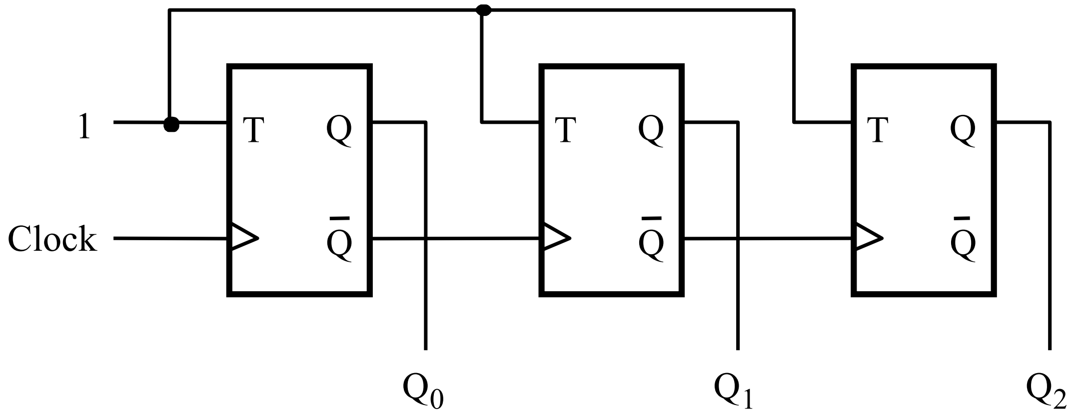
```
module counter_N_bit (clock, reset, Q);
  parameter N = 3;
  input      clock, reset;
  output     [N-1:0]     Q;
  reg        [N-1:0]     Q;
  always @  (posedge reset or posedge clock)
    if (reset == 1'b1) Q <= 0;
    else               Q <= Q + 1;
endmodule
```

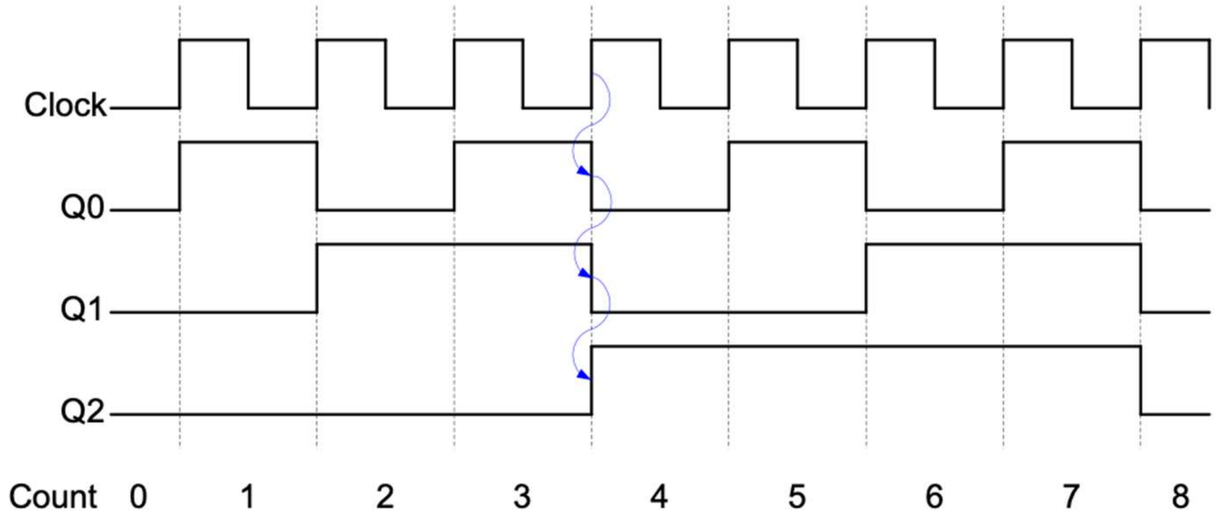
Defines a constant N



Asynchronous Binary Counter – Ripple Counter

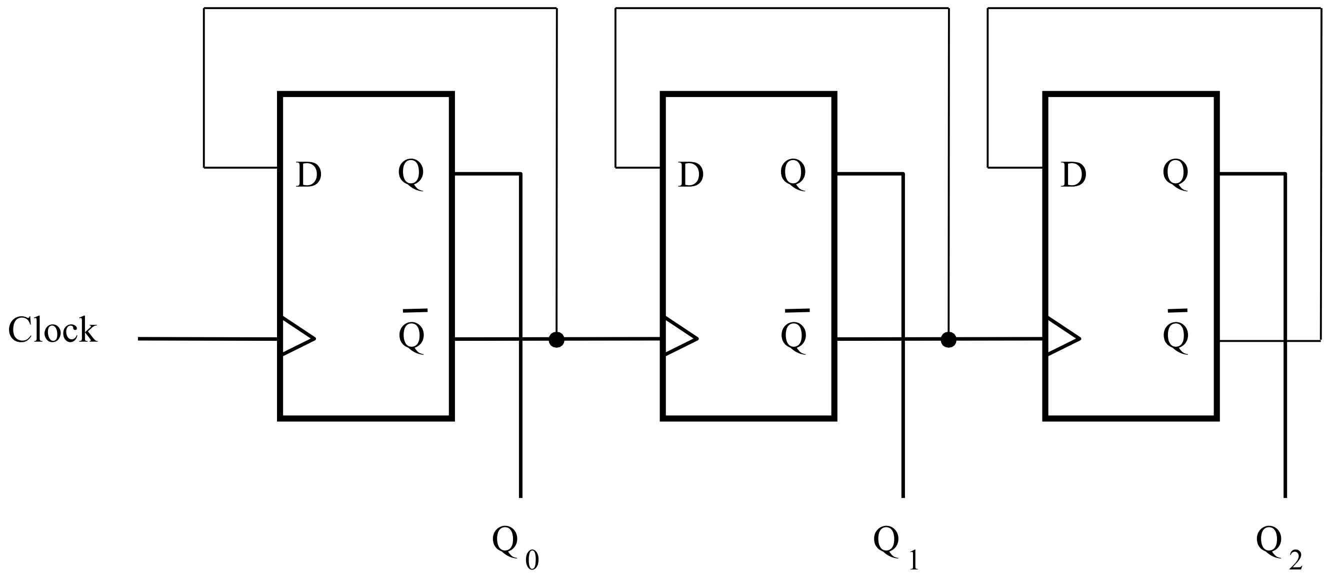
- Flip flops are not triggered by a global clock signal
  - Example: up counter with T flip flops





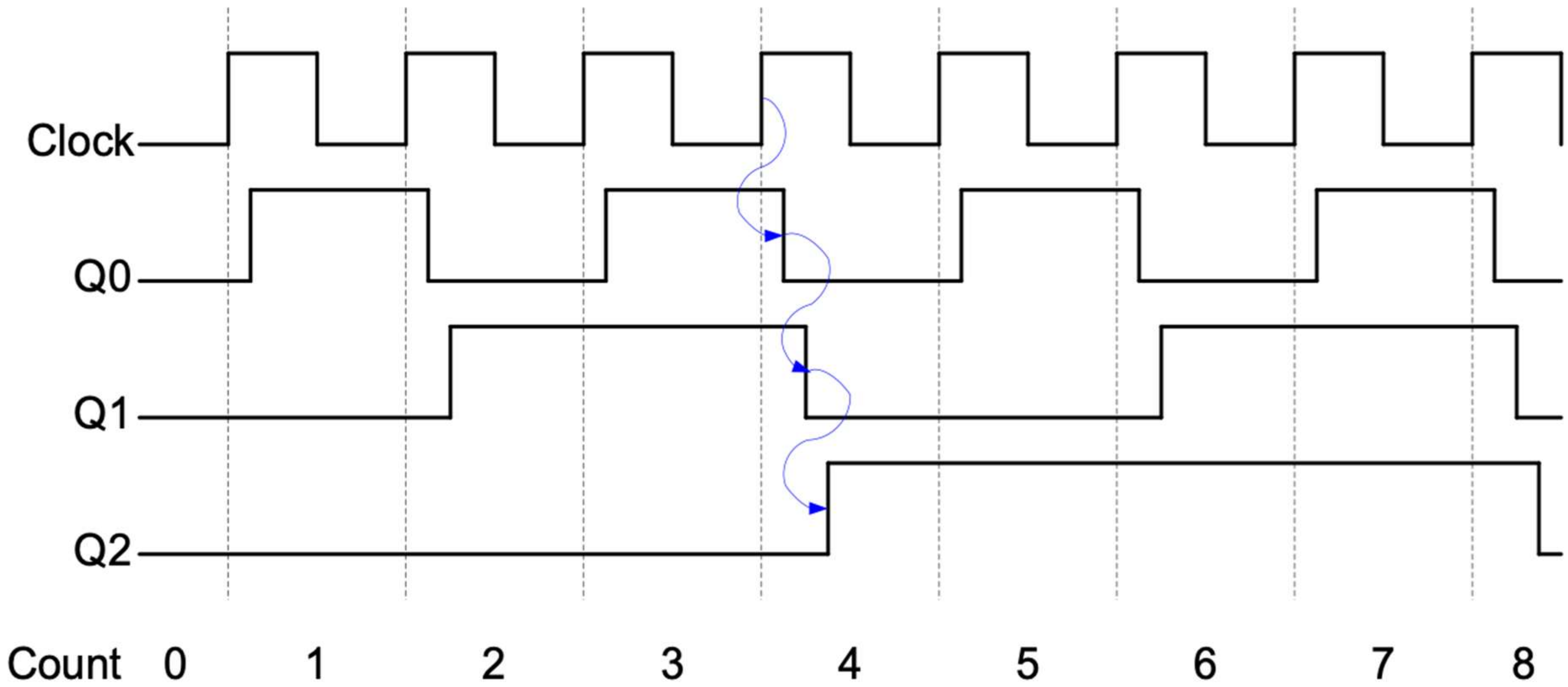
Asynchronous Binary Counter – Ripple Counter

- Example:
  - Alternative binary ripple counter, with D flip flops, up counter



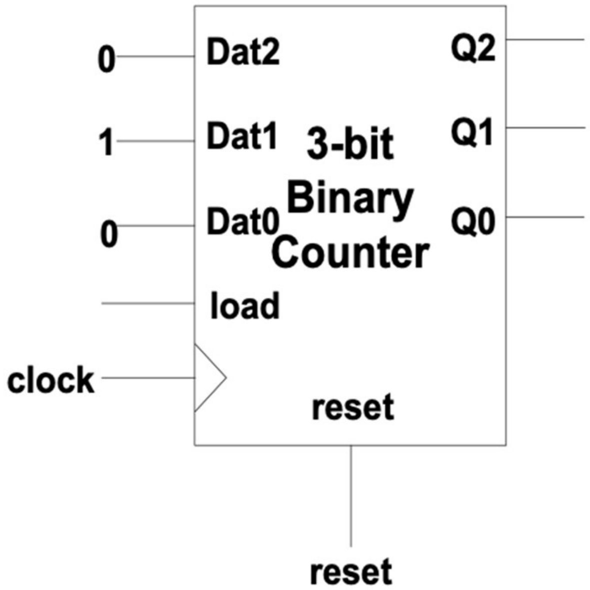
Asynchronous Binary Counter – Ripple Counter

- Problem with asynchronous counters:
  - Delays caused by each stage – timing issues



Synchronous Counter with Control Input

- Control inputs may be added to the flip flops in a binary counter to control the behavior of the counter
- **Load (Parallel Load)**:
  - Numbers can be loaded into the counter anytime when the load input is high, thus the count sequence can be customized

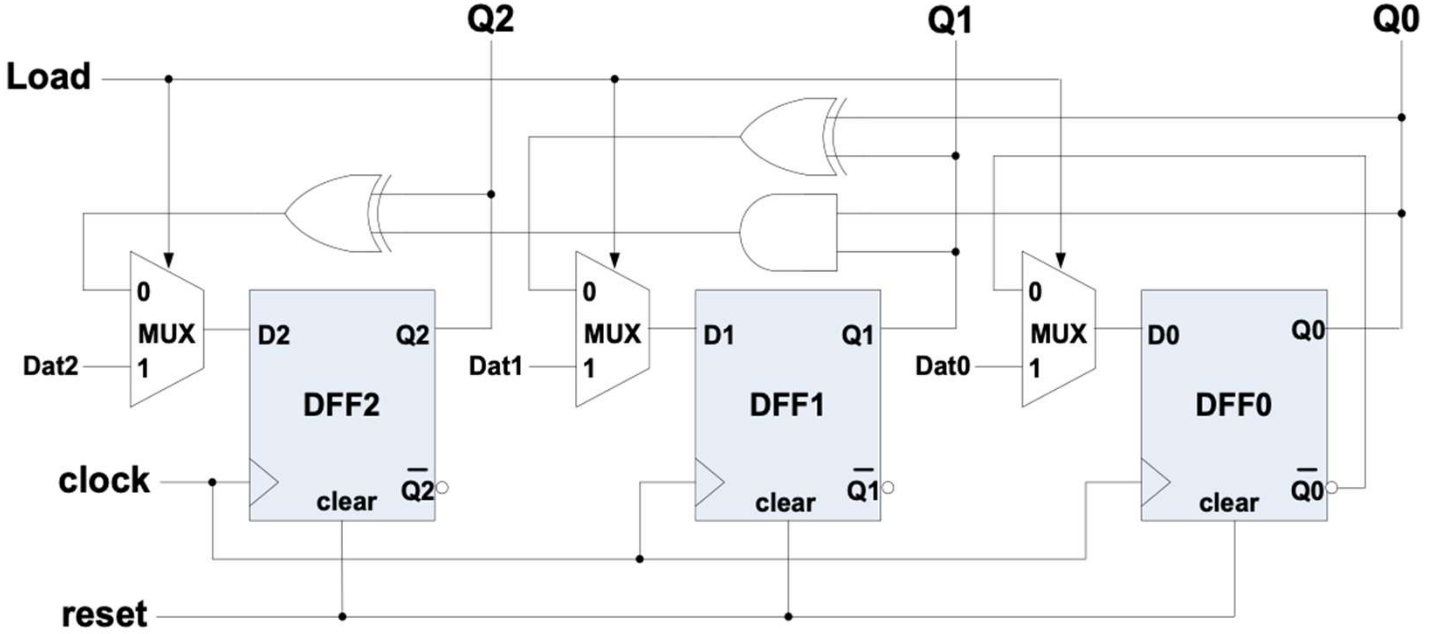


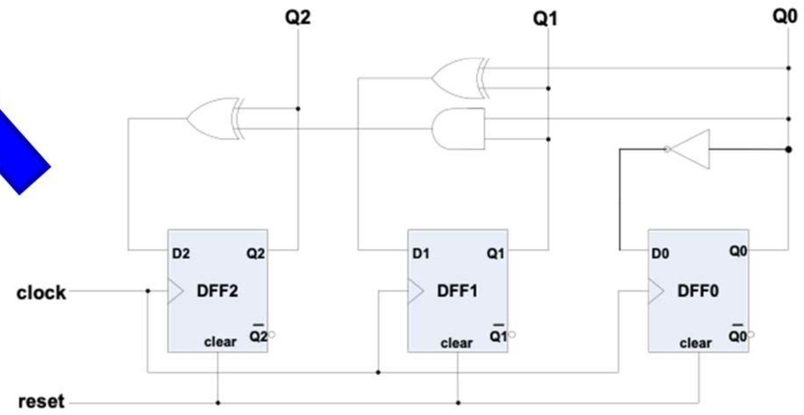
Counting sequence:

010 → … → 110 → 111 → 000 → 001 …

Implementation of Parallel Load

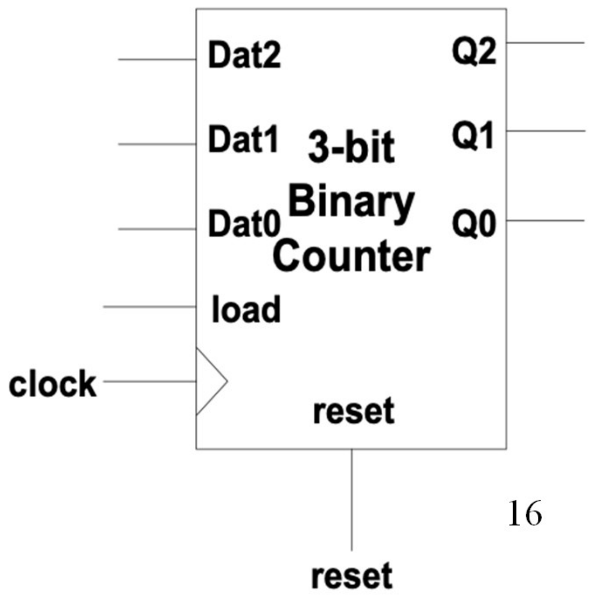
- Using ▢ for the D inputs of DFFs





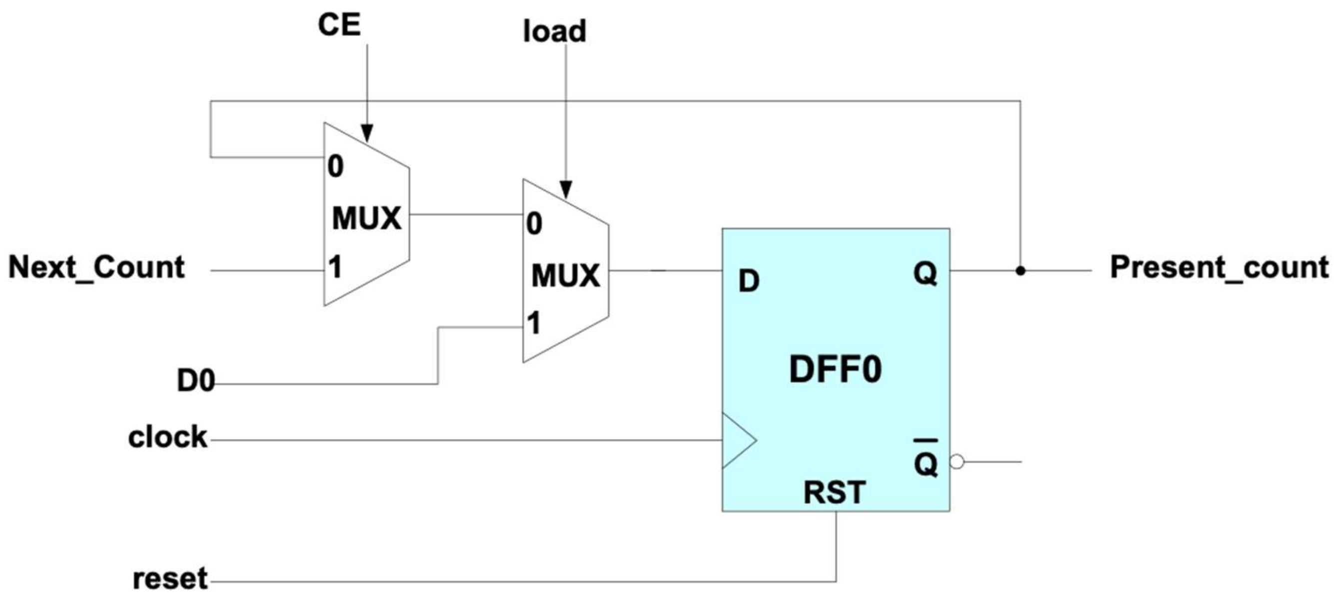
Verilog Model: Counter with Parallel Load

```
module counter_N_bit (clock, reset, load, Dat, Q);
  parameter N = 3;
  input      clock, reset, load;
  input      [N-1:0]     Dat;
  output     [N-1:0]     Q;
  reg        [N-1:0]     Q;
  always @  (posedge reset or posedge clock)
    if (reset == 1'b1) Q <= 0;
    else if (load == 1'b1) Q <= Dat;
    else Q <= Q + 1;
endmodule
```



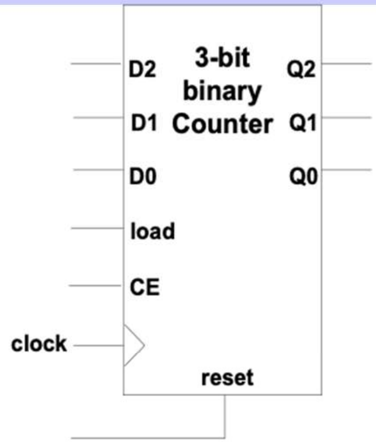
Binary Counter with Count Enable

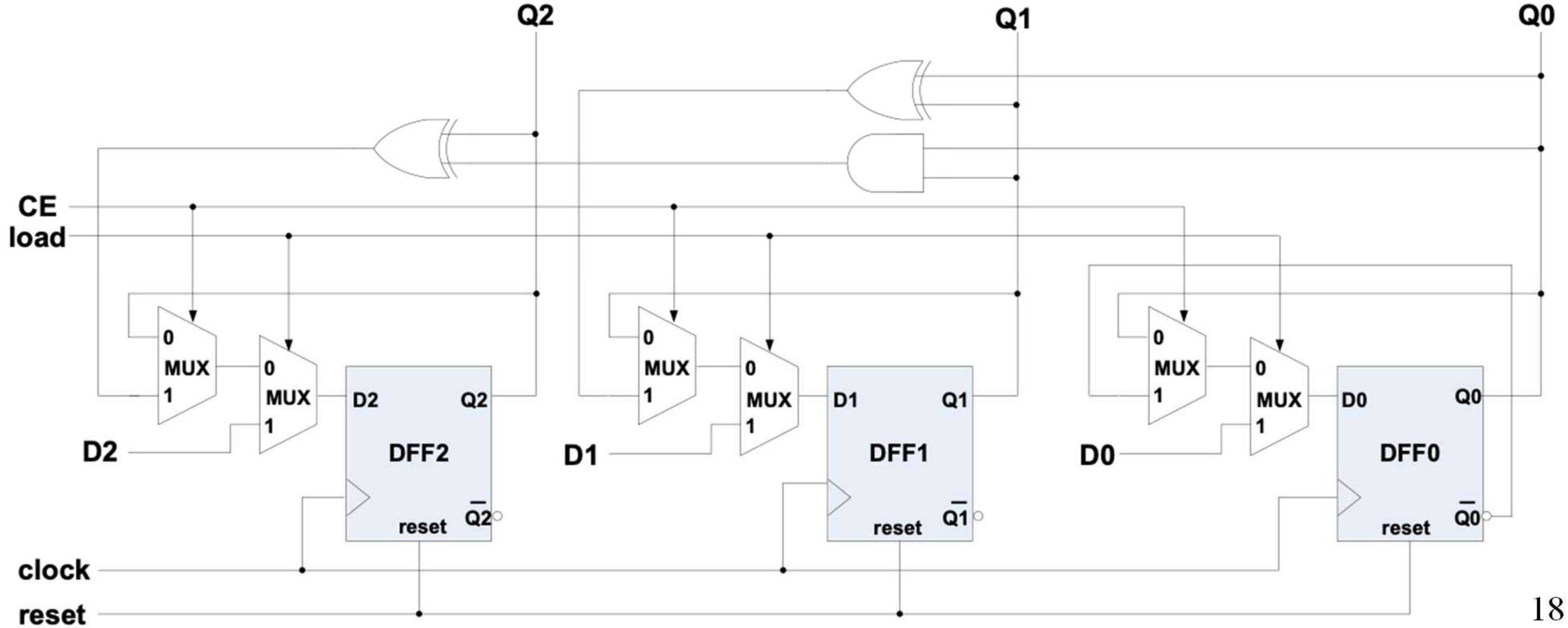
- **Count Enable (CE)** :
  - when CE = 1, counter counts
  - when CE = 0, counter holds the values
- Used to hold the counter to
  - Wait certain acknowledge signal coming from other devices
  - Concatenate small counters into bigger ones
- Implementation of 1-bit of a counter with both CE and load:



Binary Counter with CE and Load

- 3-bit binary counter with load and CE





Priority of External Control Signals

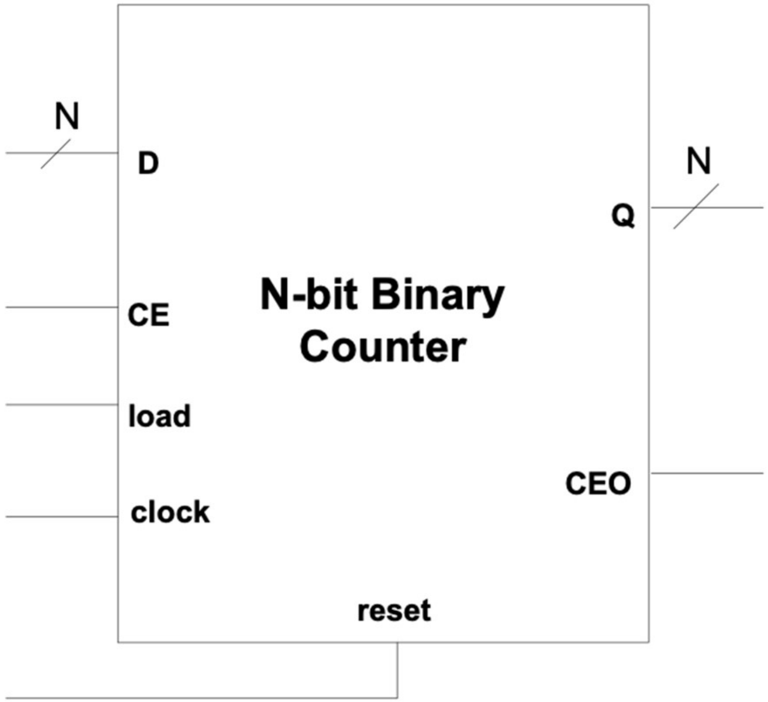
- Function table of a counter with external controls
  - a prioritized hierarchical control structure

reset	load	CE	Action on the rising clock edge
1	X	X	Clear (Qn <= 0)
0	1	X	Load (Qn <= Dn)
0	0	1	Count
0	0	0	Hold (No Change)

N-bit Binary Counter

- **Count Enable Output (CEO)**

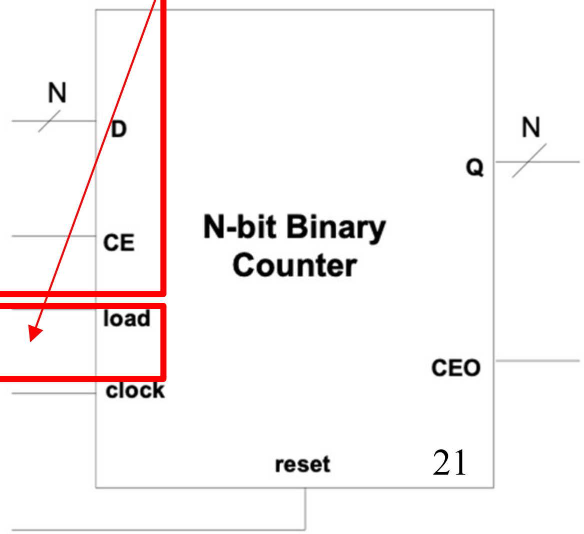
  $CEO = CE \cdot Q_{N-1} \cdot Q_{N-2} \cdot \ldots \cdot Q_0$



Verilog Model: Counter with Count Enable

```
module counter_N_bit (clock, reset, load, CE, D, Q, CEO);
  parameter N = 3;
  input       clock, reset, load, CE;
  input       [N-1:0]      D;
  output      [N-1:0]      Q;
  output                   CEO;
  reg         [N-1:0]      Q;
  always @  (posedge reset or posedge clock)
    if (reset == 1'b1) Q <= 0;
    else if (load == 1'b1) Q <= D;
    else if (CE == 1'b1) Q <= Q + 1;
    else Q <= Q;
  assign CEO = CE & (&Q);
endmodule
```

Executed concurrently

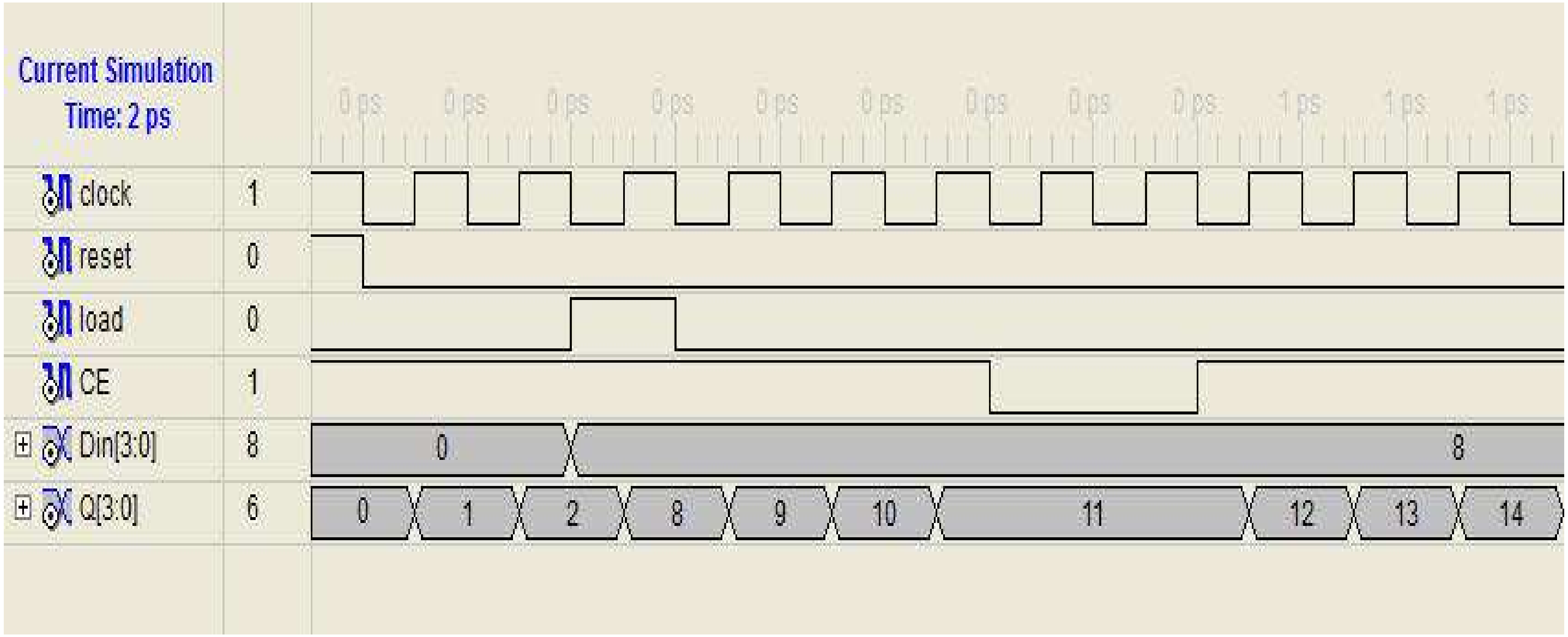


Testbench Written in Verilog

All executed concurrently

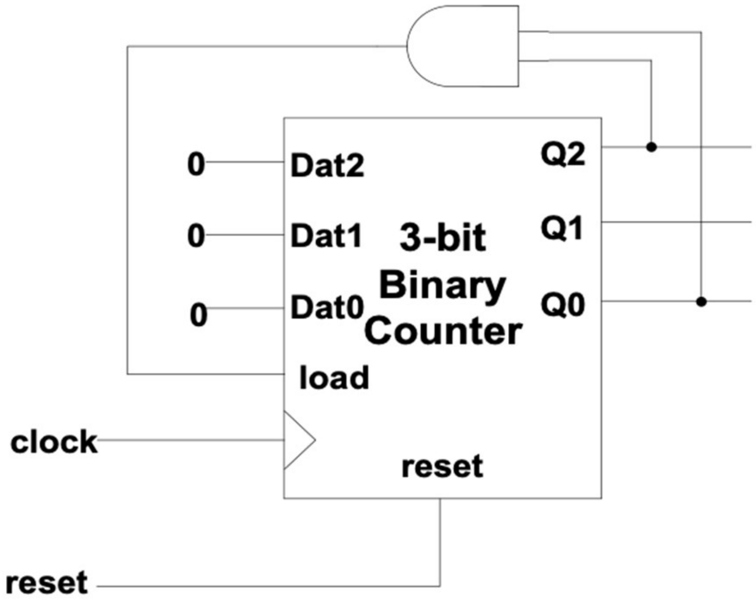
```
module Test_Banch;
  parameter half_period = 50;
  parameter counter_size = 4;

  wire [counter_size-1:0] Q;
  reg  [counter_size-1:0] Din;
  reg  clock, load, reset, CE;

  counter_N_bit #(counter_size) UUT (clock,reset,load,CE,Din,Q,CEO);

  initial begin
    #0    clock = 0; Din = 0; load = 0; CE = 1; reset = 1;
    #100  reset = 0;
    #200  Din = 8; load = 1;
    #100  load = 0;
    #300  CE = 0;
    #200  CE = 1;
  end

  always #half_period clock = ~clock;

  initial #2000 $stop;
endmodule
```

Simulation Result



Customize Counting Sequence

- What does this circuit do?

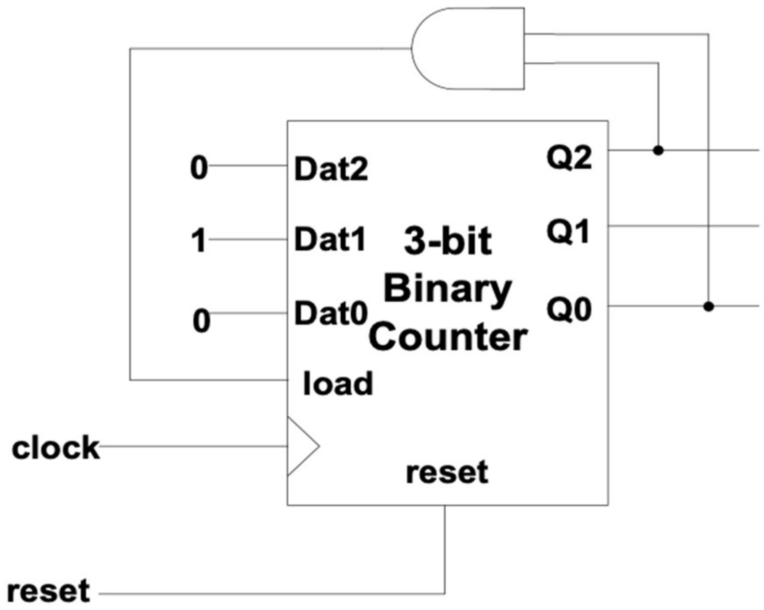


Q2	Q1	Q0	$Q2^+$	$Q1^+$	$Q0^+$
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	0	0	0
1	1	0		X	
1	1	1			

- load = Q2Q0, initial number = 000
- The count sequence now becomes
  000 → 001 → … → 101 → 000
  (modulo-6 counter)

Customize Counting Sequence

- What does this do?



Q2	Q1	Q0	$Q2^+$	$Q1^+$	$Q0^+$
0	0	0		X	
0	0	1			
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	0	1	0
1	1	0		X	
1	1	1			

- load = Q2Q0, initial number = 010
- The count sequence is

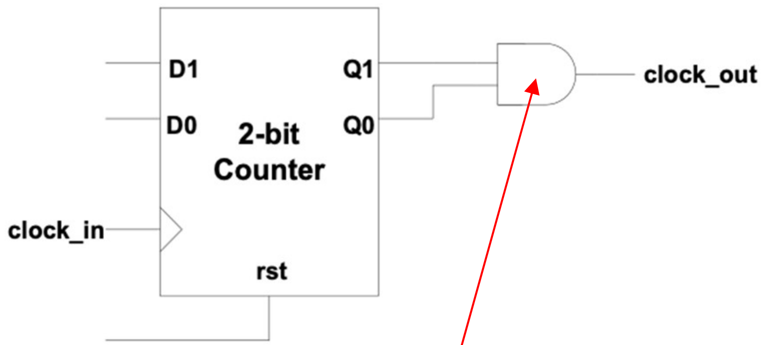
$\underbrace{\mathit{000} \rightarrow \ldots \rightarrow \mathit{101}}_{\text{In first cycle}} \rightarrow 010 \rightarrow 011 \rightarrow \ldots \rightarrow 101 \rightarrow 010 \rightarrow \ldots$

An Application of Counter: Clock Divider

- Slows a clock down by reducing its frequency
  -
  -
  -
- Generates slower pulses
  -
  -
  -

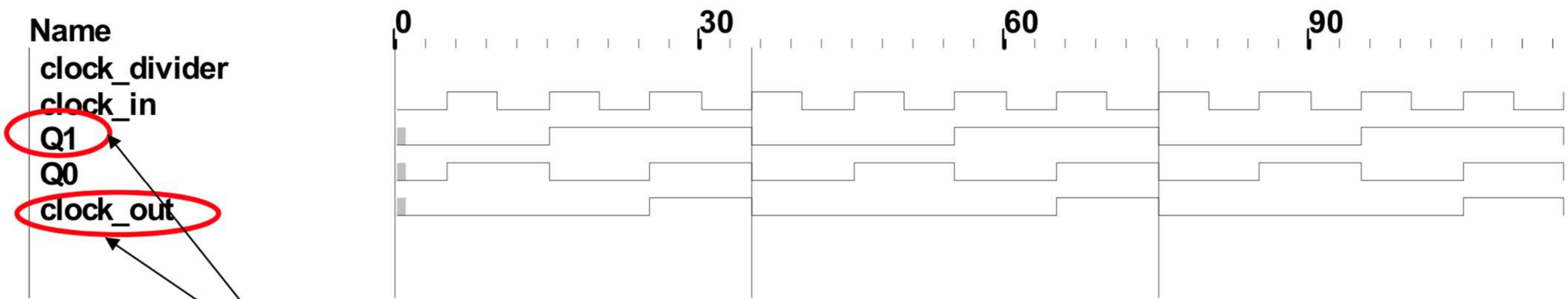
Example: Divide by 4

- 4-fold clock divider: 2-bit binary counter
- The clock_out generates one pulse for every four input clock cycles.
  - So the frequency of the clock_in is reduced by $\boxed{}$ times
- Q1 serves the same purpose with 50% duty cycle



Ideal case, delay ignored

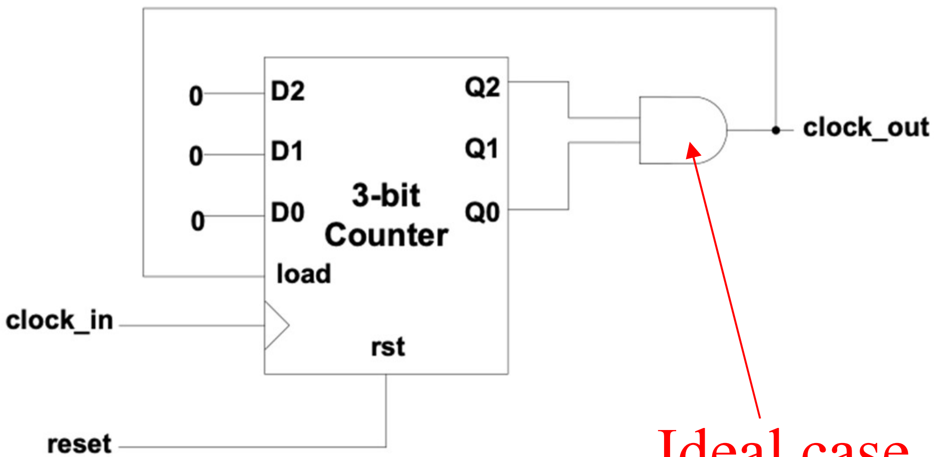
T1 35 T2 75 Tdelta 40



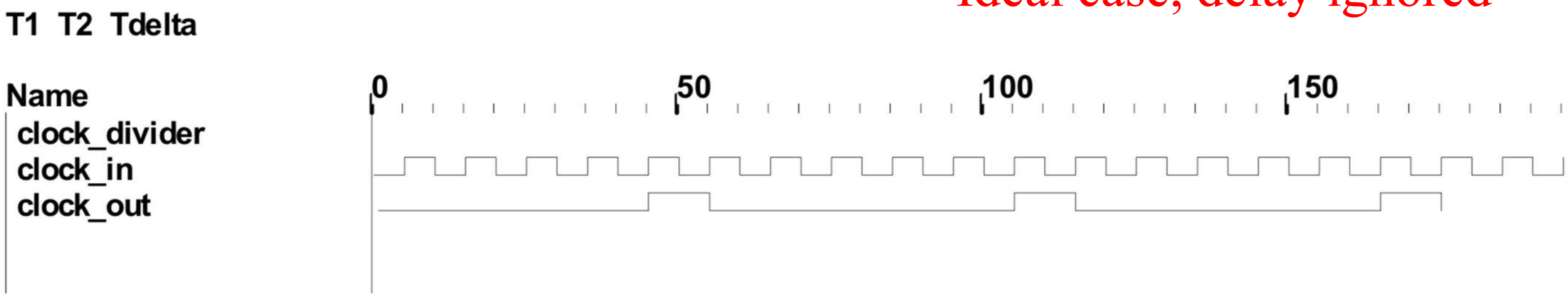
Both have the same slower frequency

Example: Divide by 6

- $2^2 < 6 < 2^3$, at least a 3-bit counter
- Countering sequence: 000→001→010→011→100→101→000
- The output should be a logic "1" whenever Q2 and Q0 are high
  - clock_out = load = Q2 & Q0



Ideal case, delay ignored



Divide by N

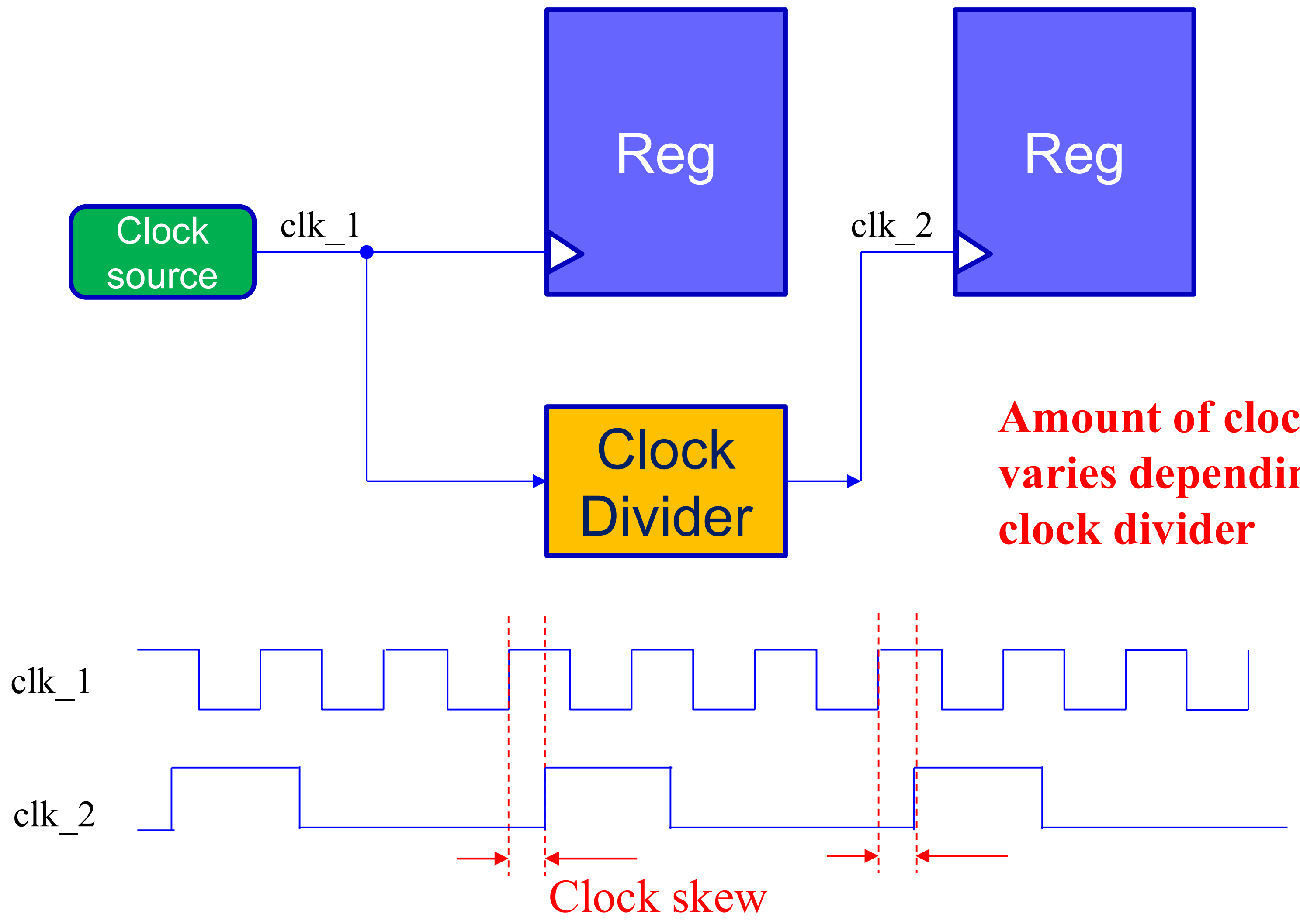
- In general, for N-fold Clock Divider
  - $2^{n-1} < N \leqq 2^n$, where n is the size of counter
  - N-1 should be the upper bound of the counter's counting sequence, i.e.

    $0 \rightarrow 1 \rightarrow \ldots\ldots \rightarrow N-1 \rightarrow 0$
  - Thus the counter should be force back to 0 using load input when it reaches the number N-1
  - If B_{N-1} is the binary equivalent of decimal number N-1, both load and output clock can be formed by ANDing the bits of '1's in B_{N-1}

Gated Clock

- Propagation delay of the output clock:
  - The output of the clock dividers comes from a combinational logic circuit (single AND gate or a net of gates in multiple levels)
    - The output will be delayed due to delay of the gates
    - Amount of delay depends on the combinational circuit
    - This kind of output is called
  - Asynchronous output may cause if the output is used as a synchronizing signal (i.e. clock)
    - **Clock skew:** the difference in the arrival time of clock signal between *two sequentially-adjacent registers* (wikipedia)

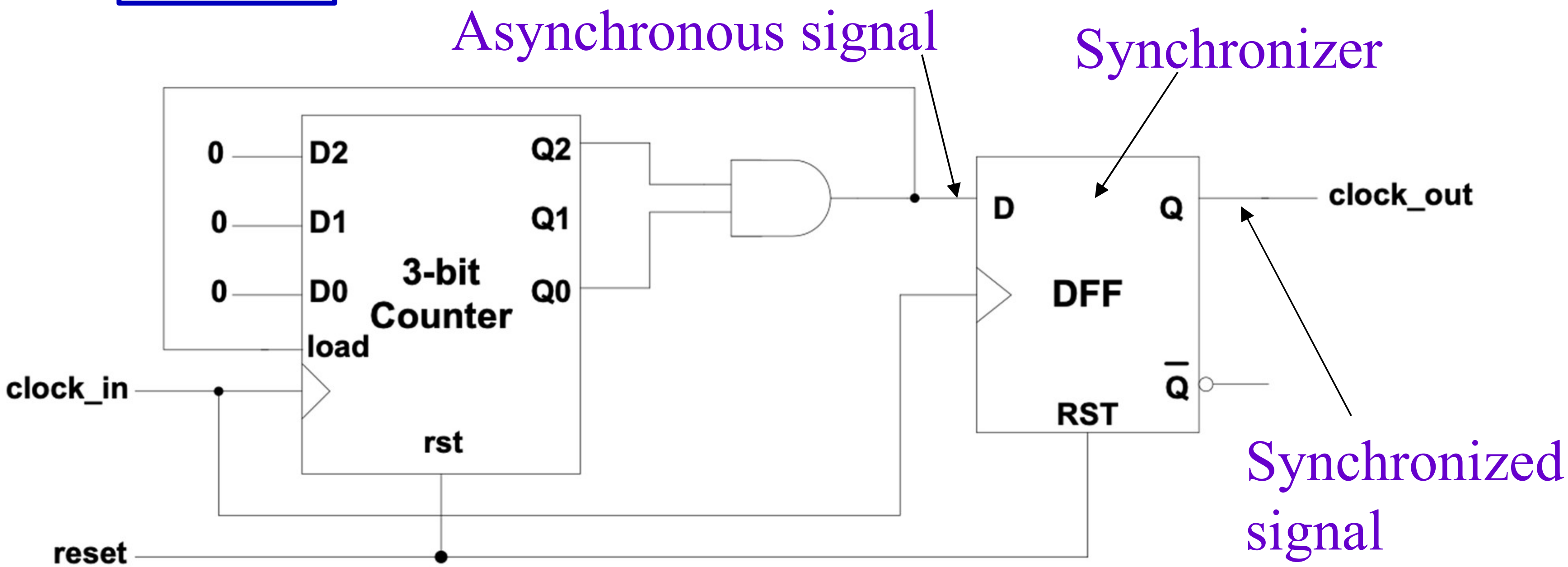
Gated Clock



Amount of clock skew varies depending on the clock divider

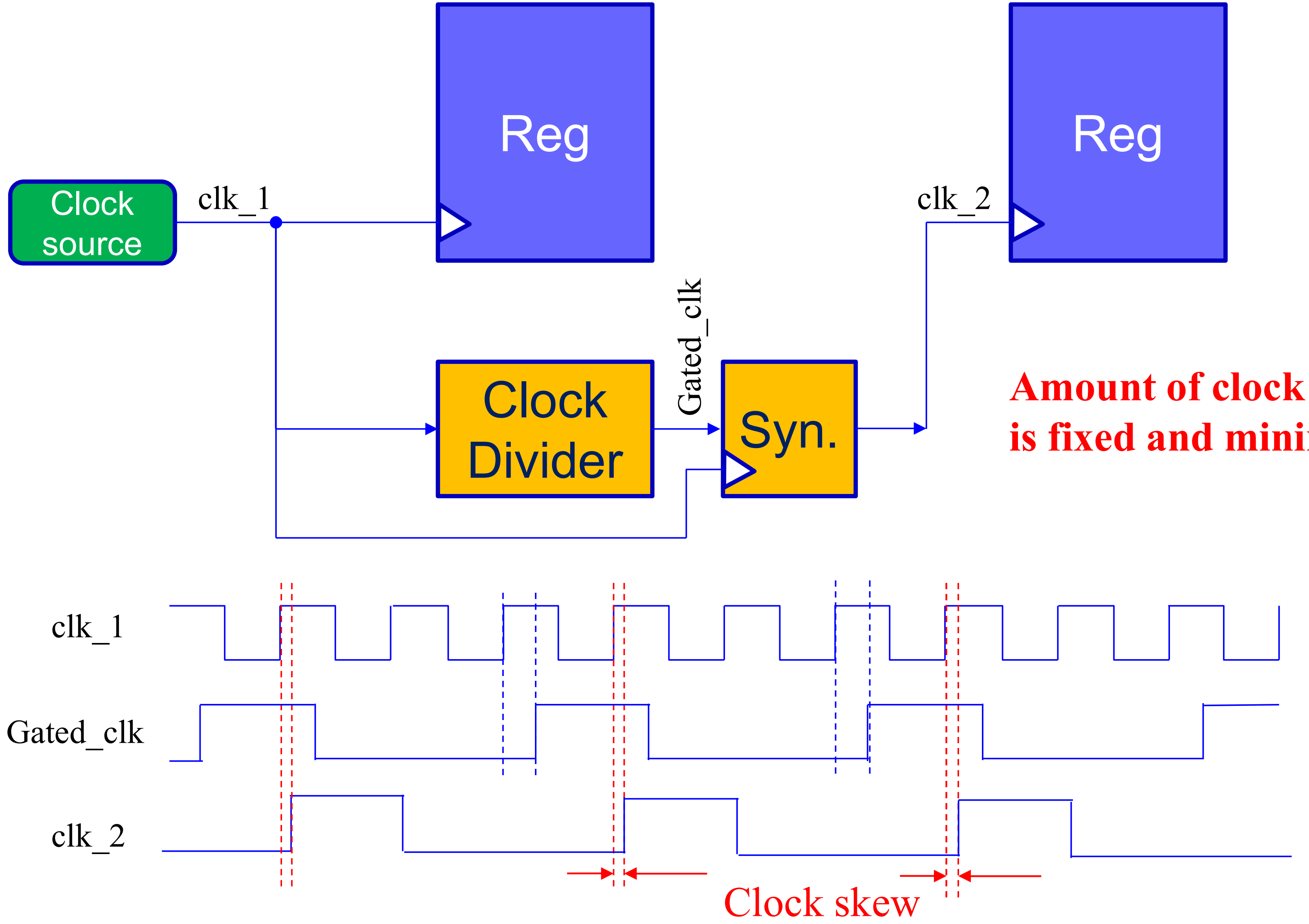
Synchronizer

- One possible solution to reduce clock skew is to synchronize the output by a



- Synchronizer delays the input asynchronous signal by up to 1 clock cycle
  - But aligns the synchronized signal with clock
  - Clock skew is not completely removed, only reduced

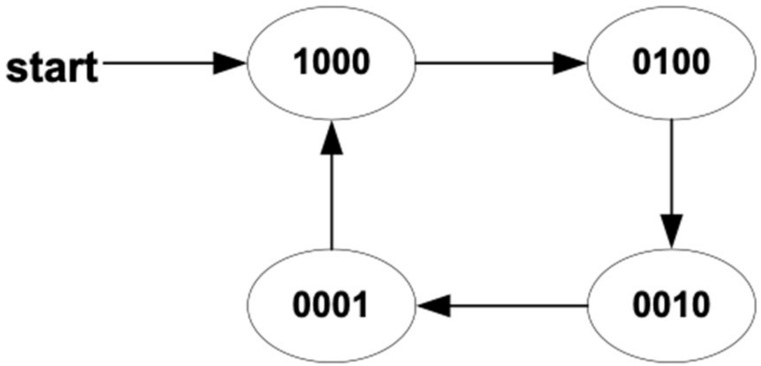
Synchronized Clock



Amount of clock skew is fixed and minimized

Ring Counter

- A ring counter is a circular shifter with only $\boxed{}$ FF being set at any time



Current				Next			
1	0	0	0	0	1	0	0
0	1	0	0	0	0	1	0
0	0	1	0	0	0	0	1
0	0	0	1	1	0	0	0
all other inputs				X			

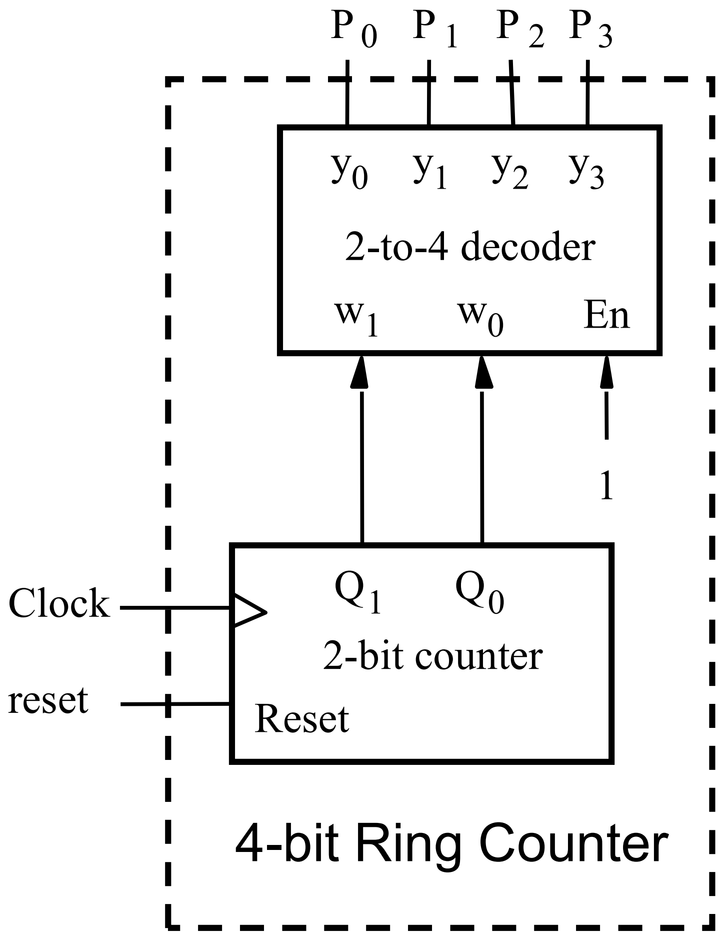
D3 = Q0
D2 = Q3
D1 = Q2
D0 = Q1

Ring Counter – Design Alternative

- A ring counter is a circular shift register with only one FF being set at any time

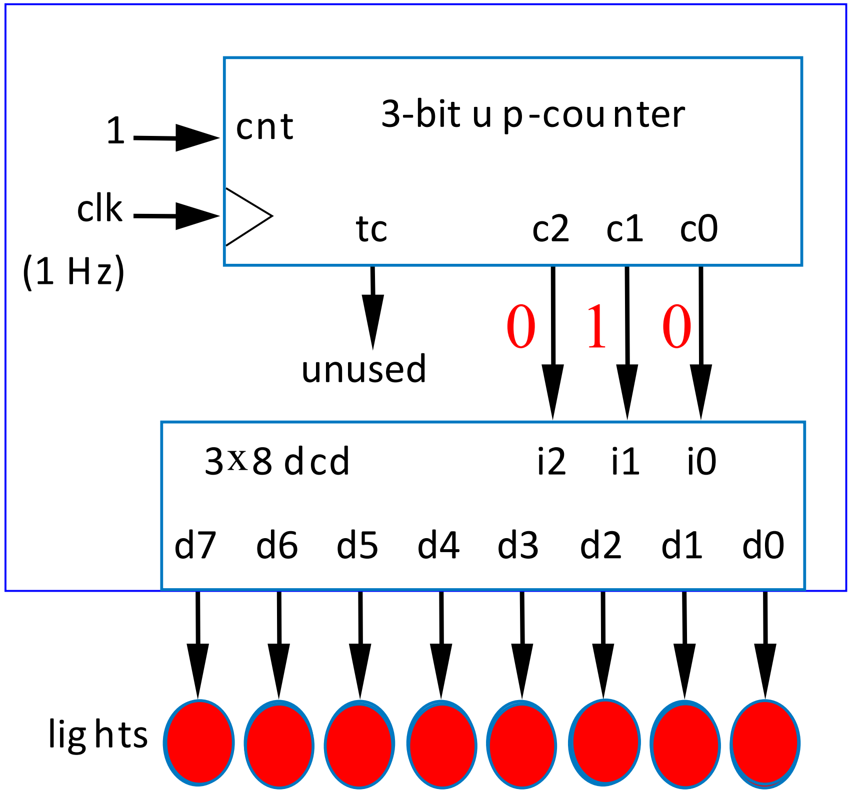
w_1	w_0	P_0	P_1	P_2	P_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

functional table of 2×4 Decoder



Ring Counter Example: Light Sequencer

- Illuminate 8 lights from right to left, one at a time, one per second
-
-



Johnson Counter

- Counting sequence

  1000 → 1100 → 1110 → 1111 → 0111 → 0011 → 0001 → 0000 → …
- In each state transition,
  - less state transition effort
  - but less counting state too

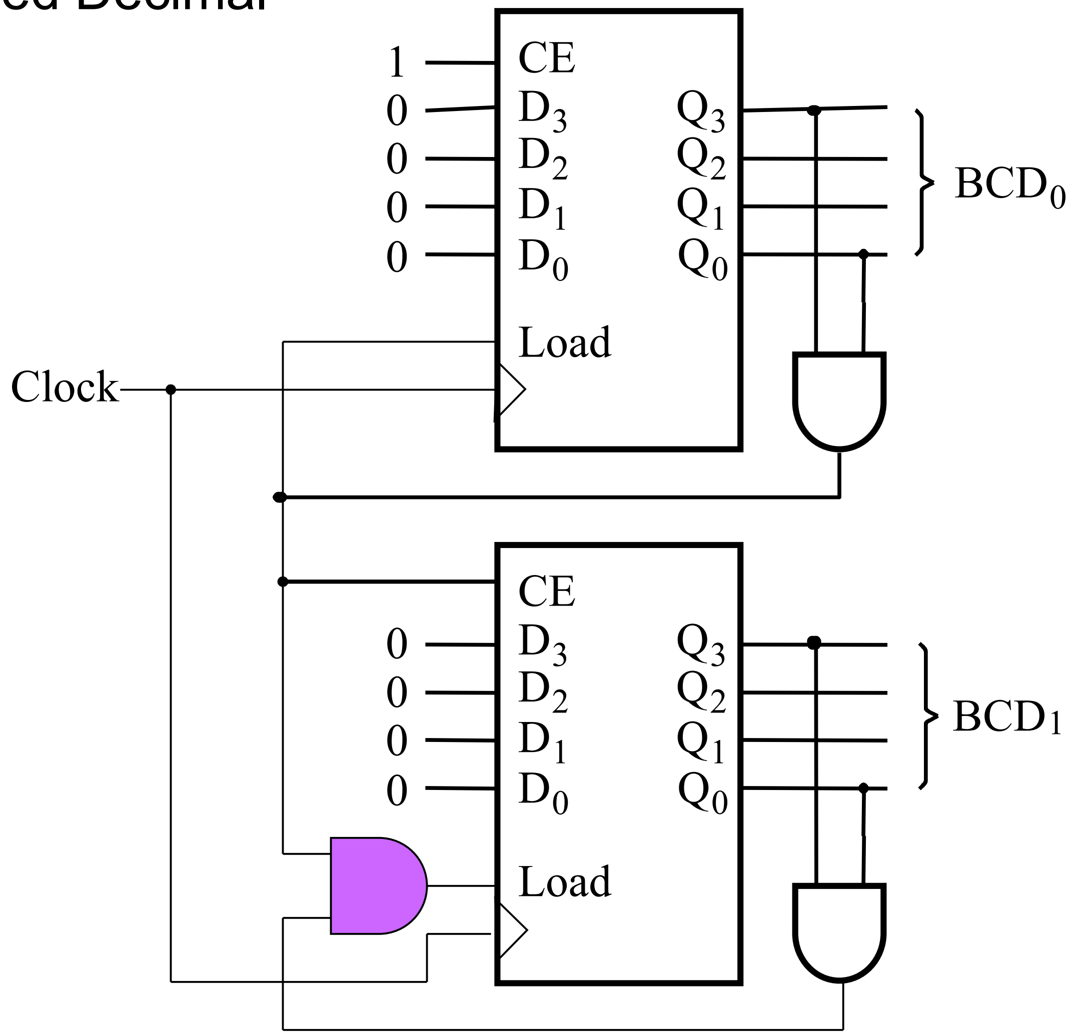
BCD Counter

- A counter doesn't have to follow through the entire counting sequence
- Example: BCD (Binary Coded Decimal) counter

Q3	Q2	Q1	Q0	$Q3^+$	$Q2^+$	$Q1^+$	$Q0^+$
0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	0	0
0	1	0	0	0	1	0	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	1	1	0	0	0
1	0	0	0	1	0	0	1
1	0	0	1	0	0	0	0
1	0	1	0				
	. . .				X		
1	1	1	1				

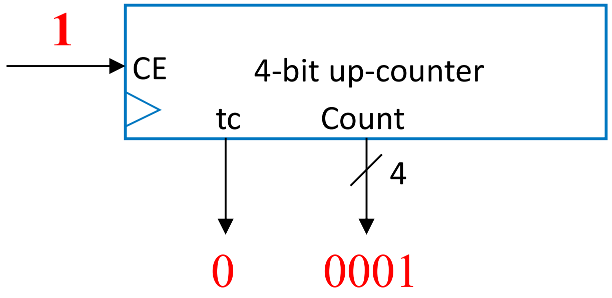
2-Digit BCD Counter

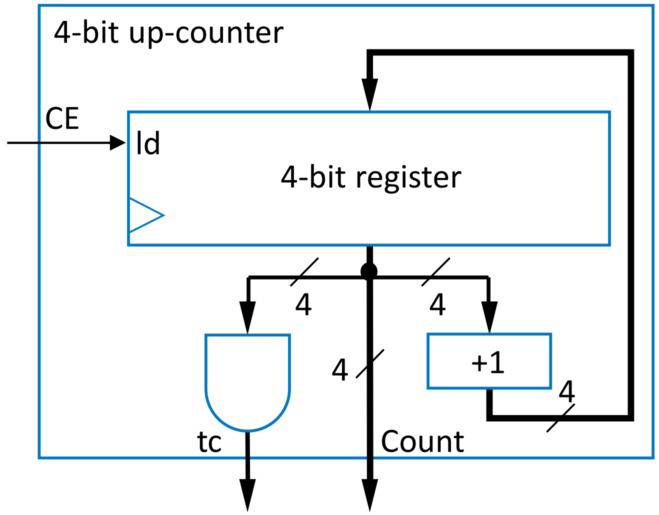
- BCD: Binary Coded Decimal



Alternative Design of Binary Counter

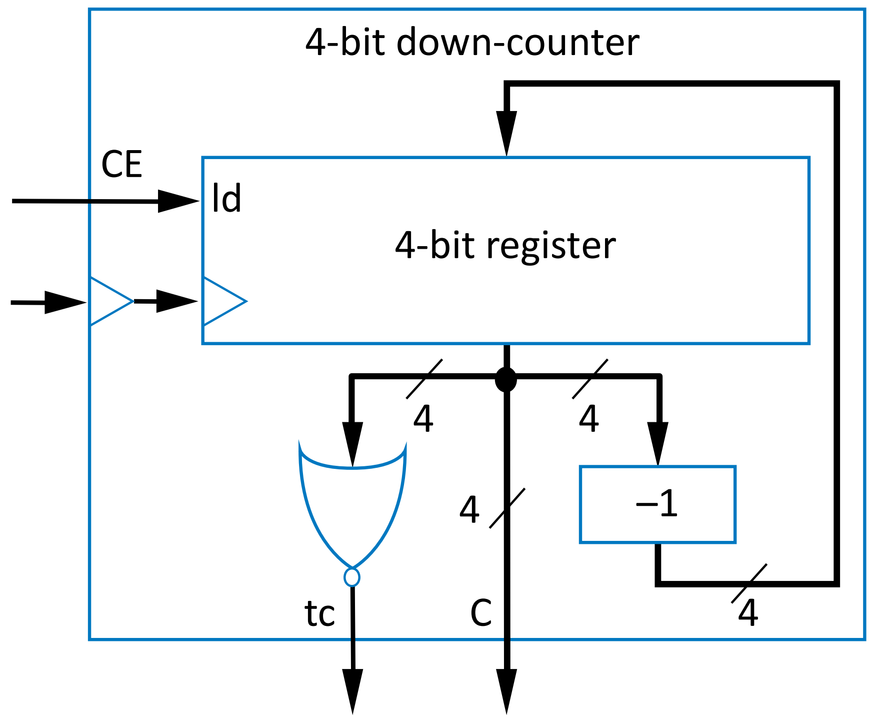
- Higher level (Register Transfer Level – RTL) design with combinational and sequential building blocks
  –
  –
  –
- Example: up counter





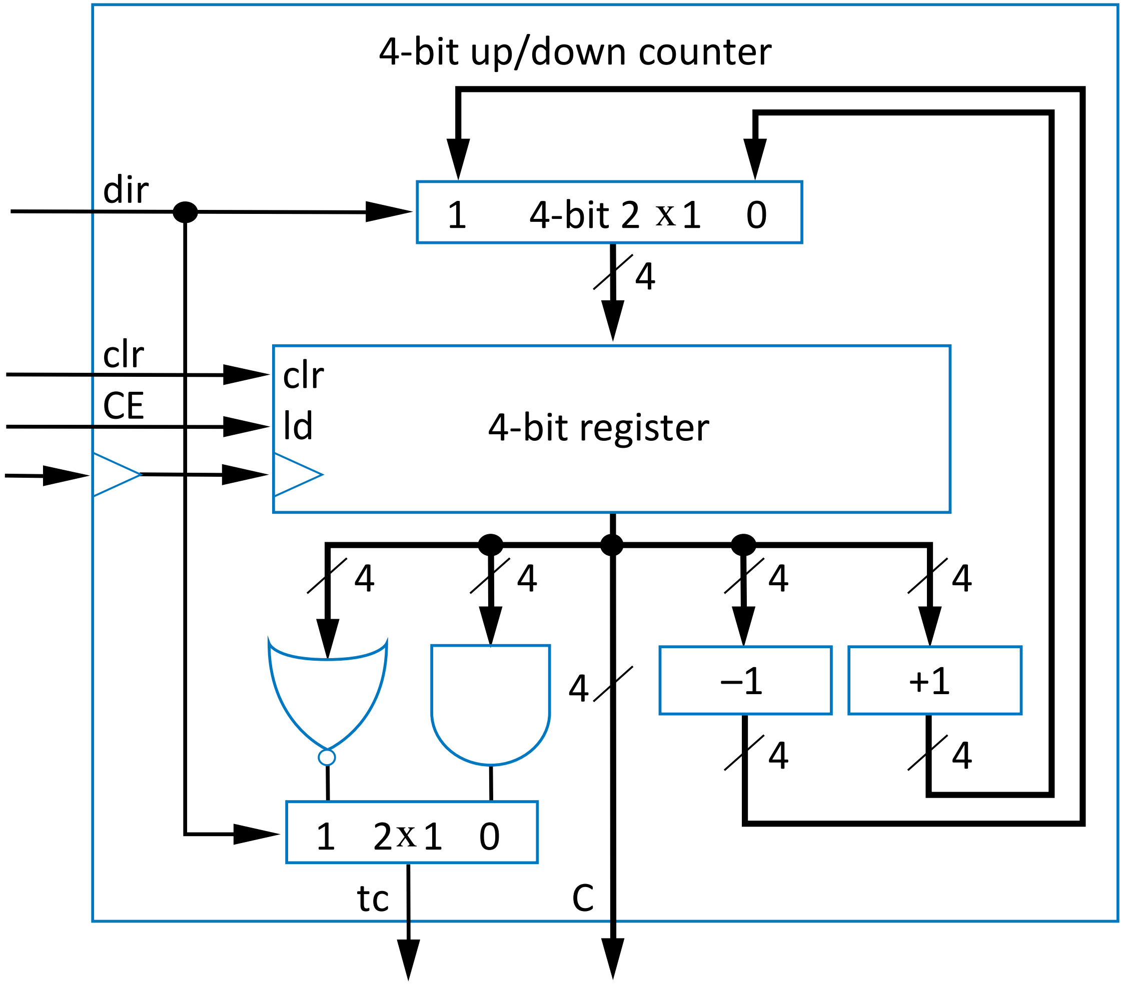
Down-Counter

- 4-bit down-counter
  - Terminal count is 0000
    - Use NOR gate for tc
  - Need decrementer (-1)



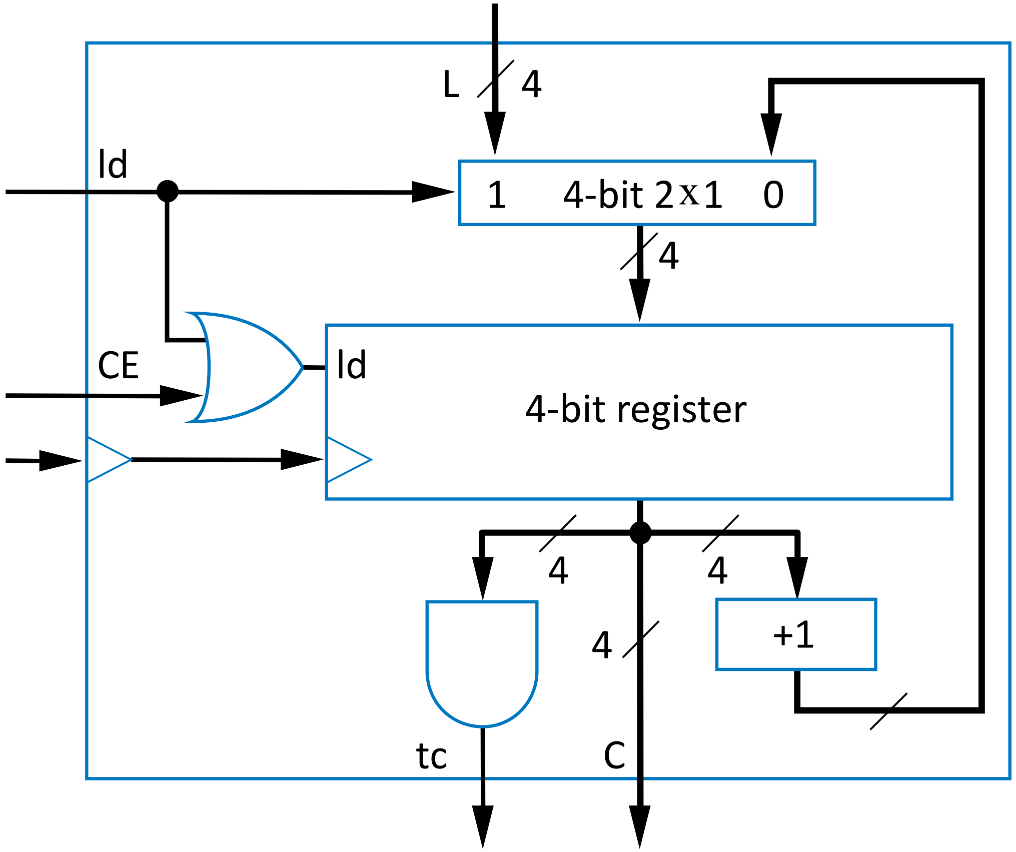
Up/Down-Counter

- Can count either up or down
  - Includes both incrementer and decrementer
  -
  -



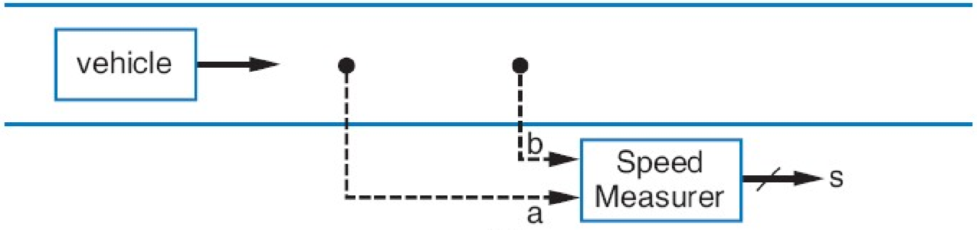
Alternative Design of Counter with Control

- Up-counter that can be loaded with external value
  - Designed using 2x1 mux – ld input selects incremented value or external value
  - Load the internal register when loading external value or when counting

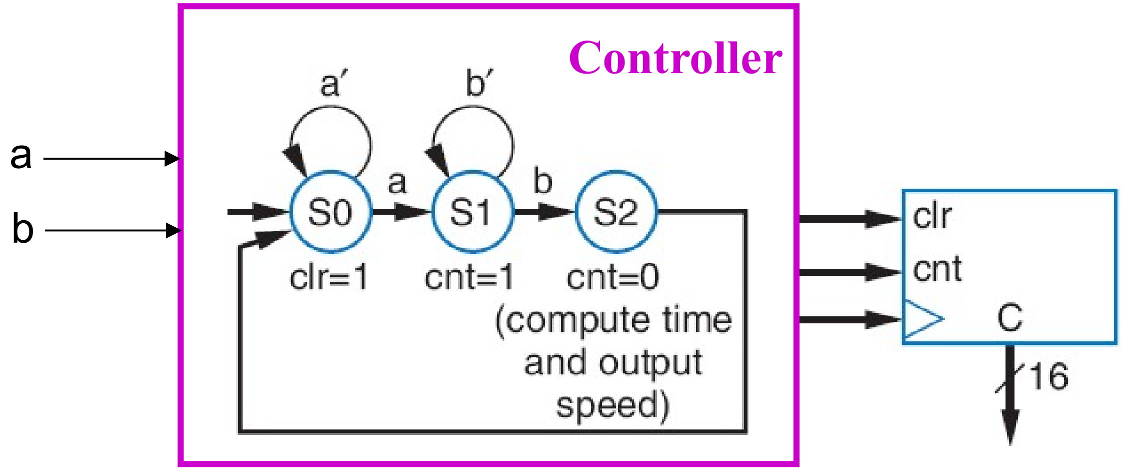


Counter Application – Timer

- A type of counter used to measure time
  - If we know the counter's clock frequency and the count, we know the time that's been counted
- Example: Compute car's speed using two sensors
  - First sensor (a) clears and starts timer, second sensor (b) stops timer
  - Assuming clock of 1kHz, timer output represents time to travel between sensors. Knowing the distance, we can compute speed



(a)



Summary

- What essentially does a counter do?
- How to design a synchronous counter w/ FFs and gates?
- Verilog modeling of counters
- How to integrate control signals into a counter (w/ MUX)?
- Test bench, test plan, and UUT
- Clock divider as an important application of counter
- What is clock skew? How to reduce it? (w/ synchronizer)
- How to put together a counter with building blocks?